DIGITAL REALTY™

Global Data Insights Survey

The State of Data and AI in **Financial Services**



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About the the survey

Goals

1. Understand how and why enterprises in financial services are rethinking data and AI strategies at the global, regional, local, and industry level

2. Explore the link between Data Gravity and enterprise strategies for storing and processing data with the goal of delivering optimal customer, employee, and partner experiences

Respondents

2,254 global IT decision-makers across **11 industries**

Respondents from **21 countries** in the Americas, Asia Pacific, and Europe

Targeted respondents in companies ranging from sub-**$100 million** small and medium-sized businesses (SMBs) to multinational corporations with over **$1 billion** in annual revenue

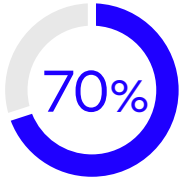


*Because of rounding, not all figures in this report add up to 100%

Key findings

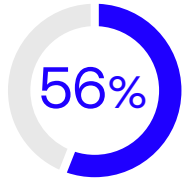
IT leaders in financial services know that to truly unlock the value of AI and reap the benefits of data-driven initiatives, they need to invest in foundational strategies that support data processing, storage, infrastructure, and even analytics tools. While they still face challenges with investment in infrastructure and leadership buy-in, they are on the right path to building a sustainable, data-centric future that will position them well for a global AI revolution.

1. Data-first strategies advance AI innovation



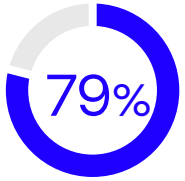
of companies in financial services are executing a formal data strategy for their current or planned IT locations. Additionally, 66% are building AI capabilities into products or services.

2. Data is pervasive and growing



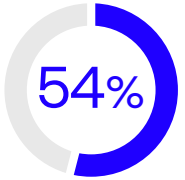
of IT leaders in financial services organizations say a lack of investing in data systems, infrastructure, and analytics tools impacts their ability to draw insights from data.

3. Data localization matters more than ever



of financial services companies follow a distributed data approach. Additionally, 70% are tying a data location strategy to their AI strategic plans so that data is located in proximity to where they do business.

4. Data drives the business agenda



Say that data-driven insights will have the biggest impact on improving customer satisfaction and experiences. These insights will help them prepare for the future of AI.

About AI maturity

In the Global Data Insights Survey, we calculated AI maturity based on self-assessed ratings on a scale from 1 to 5:

1 = No AI utilization

5 = AI is transformative/a key part of our business model

Out of the more than 2200 IT leaders we surveyed, those who who rated themselves "5" on the ratings scale **and** have a distributed data strategy are referred to throughout this report as **Data and AI Leaders**.

The Data and AI leaders provide a global benchmark for AI maturity. In this report, we will make comparisons to how these leaders are approaching data and AI compared to the financial services industry.

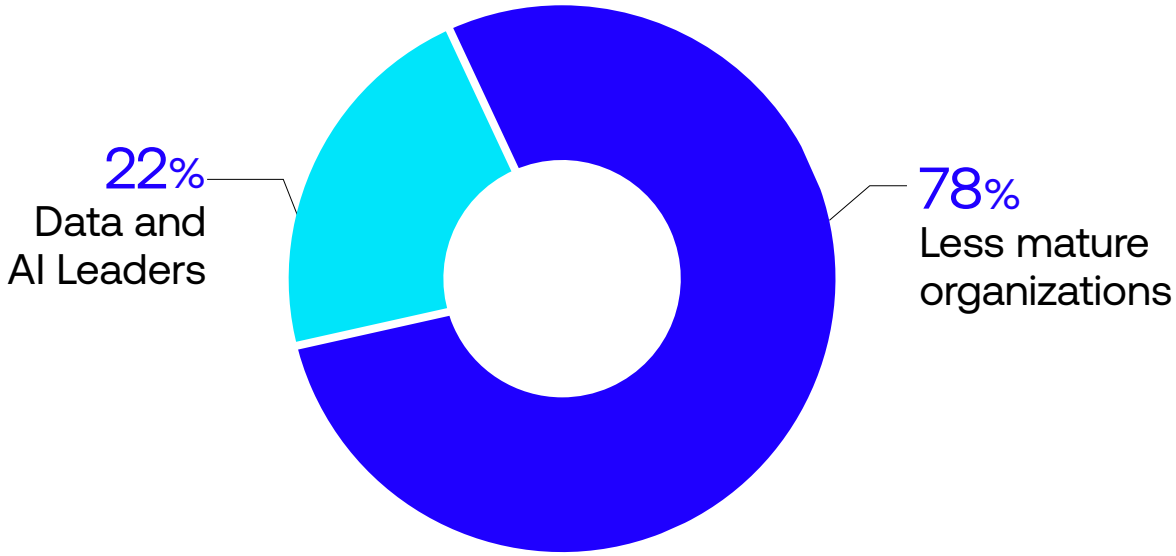


Figure 1: Digital Realty, *Global Data Insights Survey*, August 2024

A playbook for AI success

Data and AI Leaders are operating from a common playbook. This includes:

- Localizing data storage and compute resources that are seamlessly connected and highly calibrated in support of a distributed enterprise
- Leveraging formal data and AI strategies that serve as guideposts for IT infrastructure investments
- Prioritizing cybersecurity, governance, and compliance concerns at every stage of AI workload deployment

Data Gravity: A growing challenge and opportunity

Data Gravity refers to the idea that data has mass. As that mass grows, it becomes increasingly difficult to move or replicate.

Data Gravity is the attractive force caused by enterprise data creation and exchange, drawing from applications, servers, and other sources. As a data set grows, it attracts applications and services, creating a virtuous cycle of more data creation.

Data growth attracts applications and services

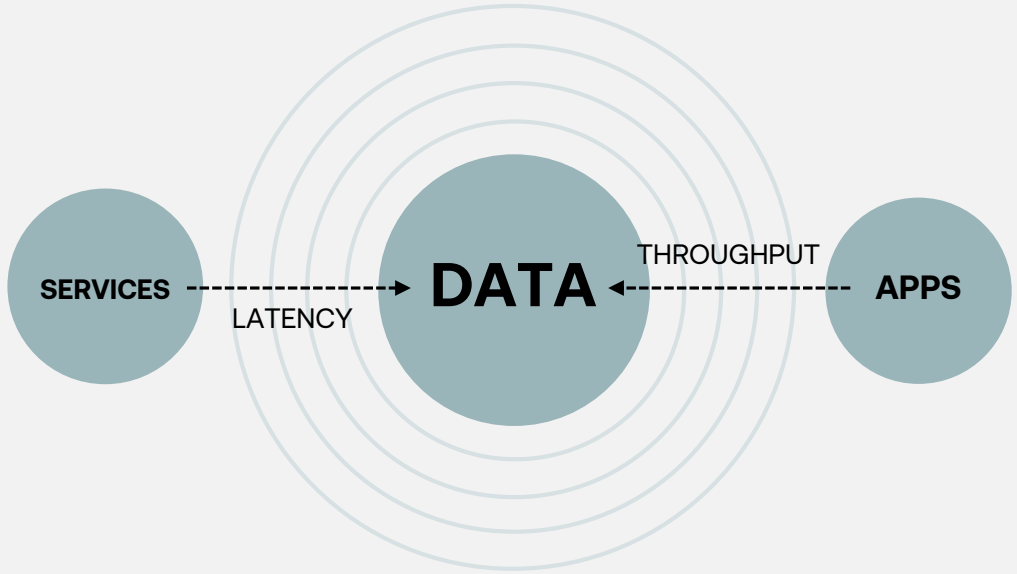


Figure 2: Patented Data Gravity Formula, McCrory and Bishop, Digital Realty

Data and AI readiness in financial services



Organizations today generate more data than ever before. Having access to that data means having more visibility into operational efficiencies, creating more personalized products and services for customers, and making decisions based not on gut or assumption, but on facts and information. Data also fuels transformational AI initiatives.

Yet alongside these opportunities are challenges. Enterprises need strategies for managing and storing data, which include meeting infrastructure needs that allow performant access to that data. Organizations also need to ensure that their data is secure and that it complies with local and regional regulations — especially in the highly regulated financial services industry. This proves a challenge in regions like Europe, where data privacy regulations like the General Data Protection Regulation (GDPR) dictate how data is collected and stored.

Another challenge is Data Gravity. **As data collects in one location, it becomes more difficult to move that data away from its current location**. Yet taking a distributed approach to data offers benefits to extract insights and opens up new opportunities.

As AI initiatives begin to be woven into the fabric of financial services enterprises, IT leaders need to shore up the strategies they have, or adopt new ones that will better position them to maximize emerging technologies into the future. Here's how financial services companies are building their data strategies.

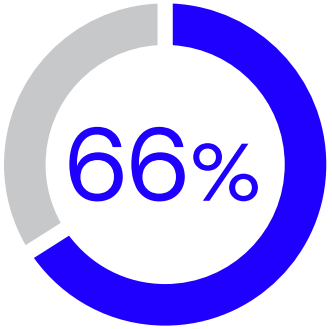


Trend 1

Data-first strategies advance AI innovation

Enterprises that rely on data to guide their decision-making are better positioned for increased innovation, earnings, and growth. Giving a boost to that growth is AI adoption. Deloitte finds that of companies using generative AI, 48% are uncovering new insights with AI, 55% are reducing costs with AI, and 63% say generative AI is encouraging innovation and growth.[1]

This is why our survey respondents say that AI and machine learning (ML) technology are the best way to extract insights from their organizational data, and that investing in these technologies will better position them to increase how they use data to drive decision-making.



are building AI capabilities into product or services

[1] Deloitte, *Deloitte's State of Generative AI in the Enterprise*, April 2024.

IT location planning is paramount to data and AI strategies

Every data-driven initiative — especially AI — needs to be built upon a strong data strategy that considers data governance, sovereignty, and infrastructure. This is why 70% of respondents in financial services have a formal data strategy that specifies the requirements for current or planned IT locations. **This strategy ensures that data is being stored in locations that make sense for its users, that the proper hardware is in place to support the high energy needs AI requires, and that locations follow the regulations unique to its region.**

Part of that strategy needs to include considerations around sustainable energy use as well, especially when **93% say sustainability has an impact on their AI strategy**.

However, there is room for improvement, as those in financial services are lagging behind 79% of global Data and AI Leaders who say they are actively executing their formal data strategy for their IT locations.

Does your company have or plan to have a formal data strategy specifying the requirements for current or planned IT locations?

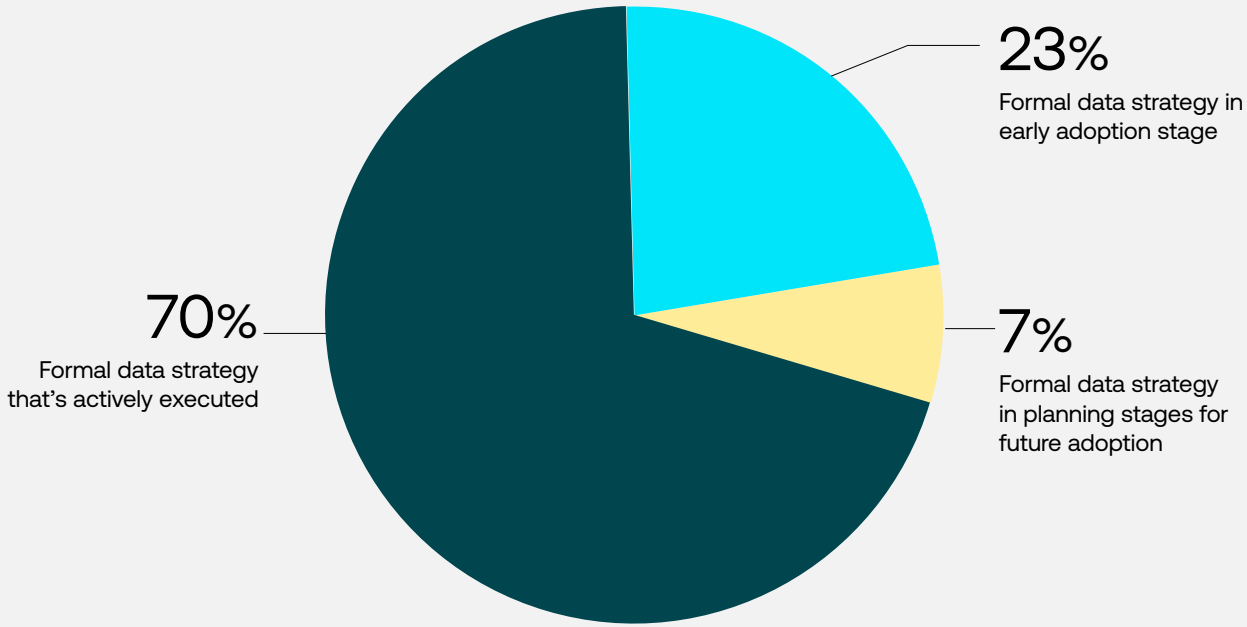


Figure 3: Digital Realty, *Global Data Insights Survey*, August 2024



Trend 2

Data is pervasive and growing

IDC predicts that unstructured data will grow at an annual rate of 21.2%, hitting 221,000 exabytes by 2026.[1] Because of this, financial services organizations need to standardize their strategies for AI and data-driven initiatives because the amount of data they need to manage will only continue to grow.

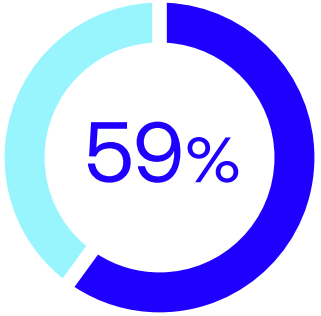
IT leaders in financial services are feeling the pain points of Data Gravity. **The biggest obstacle they face when drawing insights from their data is a lack of sufficient investment in data systems, infrastructure, and analytics tools, experienced by over half (56%) of respondents**. This not only signals a need for a more comprehensive data strategy, but also the need for more budgetary allotment and commitment from leadership to fund those needed investments.

When it comes to supporting their AI initiatives, six out of ten (59%) say that a lack of IT infrastructure in the right locations is the primary hindrance to their AI strategy development. This can result in increased latency or not having the right energy support to facilitate AI applications.

[1] IDC, *Worldwide Global DataSphere and Global StorageSphere Structured and Unstructured Data Forecast*, 2022-2026.

Additionally, 64% say that a lack of data storage required to house massive data sets AI requires is the biggest hurdle to adopting an AI strategy — which is slightly more than 60% of global respondents, highlighting the need for data storage financial service companies.

To be better positioned to draw insights from data, organizations need to invest in data systems, infrastructure, and analytics tools, as well as infrastructure that can handle AI's power and energy demands to store, process, and host vast quantities of data.



say lack of IT infrastructure in the right locations is the primary challenge to their AI strategy development

What are the top three obstacles your organization faces when drawing insights from data?

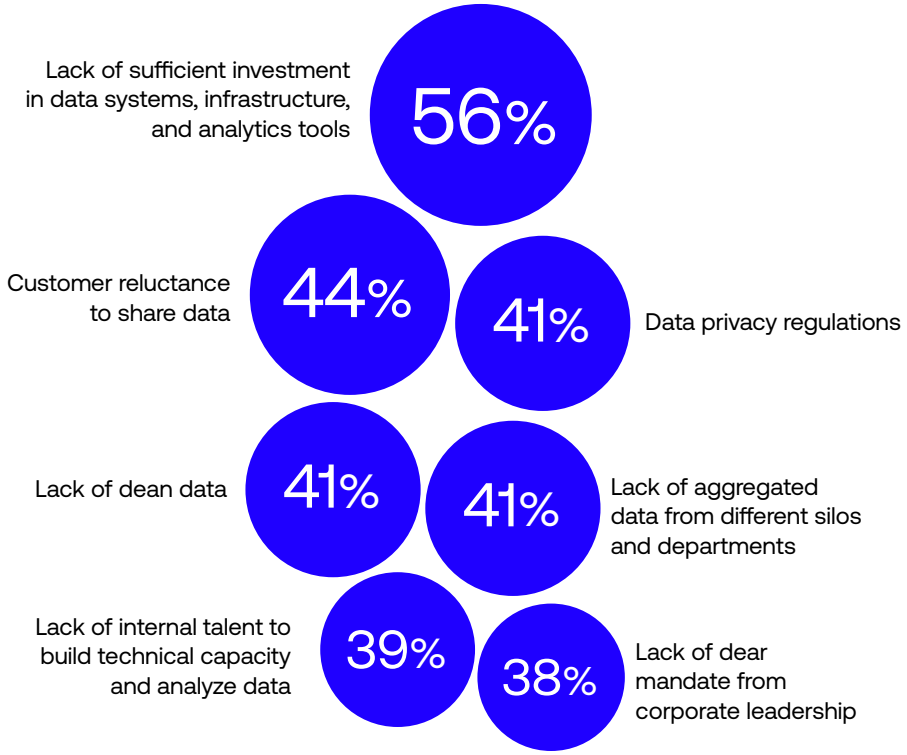


Figure 4: Digital Realty, *Global Data Insights Survey*, August 2024



Trend 3

Data localization matters more than ever

Financial services companies also need strategies to manage where their data is being collected and stored, and how that data complies with privacy regulations and compliance in varied regions globally.

This is why data localization is becoming more of a priority for IT leaders. Not only do IT locations need to have the right hardware to support emerging technology initiatives like AI, IT leaders need to create plans that align with local and regional regulations. They also need to consider data security: who has access to data, how access is controlled, and how networks and systems — including ever-expanding cloud environments — are secured.

Taking a distributed data approach allows data to live closer to those who use it, making access and analysis faster and more efficient.

"If you don't have the right data where you need it, then your AI strategy is broken before it starts. Data must be in the right place where AI can ingest it and create more data in a forever perpetuating cycle."

Dan Eline
Vice President, Platform Solutions, Digital Realty

Future IT investments should account for Data Gravity

70% of respondents are currently executing a formal data strategy that specifies the requirements for current or planned IT locations, meaning that they're prioritizing location as they plan for digital transformation. **However, 59% say that a lack of IT infrastructure in the right locations is the primary hindrance to their AI strategy developmen**t — and that having the right IT infrastructure in the right locations will accelerate AI development.

Additionally, 70% say they're tying a data location strategy to their AI strategic plans — much fewer than 84% of Data and AI Leaders worldwide who are doing so. Doing so will ensure that data is collected, managed, and stored appropriately, meeting local and regional regulations. This also means that data can be located in proximity to where AI projects and deployments will launch.

As they plan ahead, IT leaders need to ensure that future investments account for localization needs, as well as support the energy requirements for AI in sustainable ways. **In planning for future investments, respondents say that what they need the most from those locations are specified latency or performance requirements for their data processing**. They also require the ability to maintain local copies of applications, workflows, and customer transactions, further supporting their localization efforts.

Are you tying a data location strategy to your AI strategic plans (so data is located in proximity to your users and locations where you do business)?

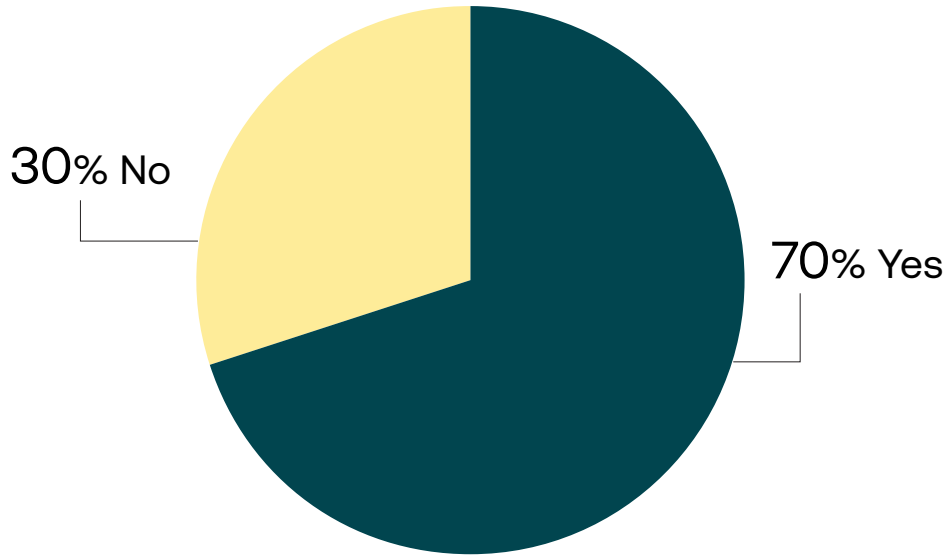


Figure 5: Digital Realty, *Global Data Insights Survey*, August 2024

Financials services leaders find value in distributed data strategies

Data localization considerations come into play when organizations take a distributed approach to collecting, storing, and processing their data — something that 79% of financial services companies are already perusing, and that's comparable to other companies worldwide.

65% of respondents say their companies have deployed IT infrastructure across one to ten locations globally, with 77% planning to expand to upwards of five additional locations in the next two years. Additionally, 29% have deployed AI projects across more than half of their locations.

But why take a distributed approach to data? Financial services IT leaders selected two reasons that tied for the top slot.

1. It allows data to live closest to those who most need it (44%), thereby making access to data faster and easier.
2. It allows them to draw richer insights from their data as well (44%).

Ultimately, latency requirements are at the heart of IT location strategy — requirements that IT leaders need to ensure their data center service providers can meet.

What are the top 3 reasons your company follows a distributed approach?

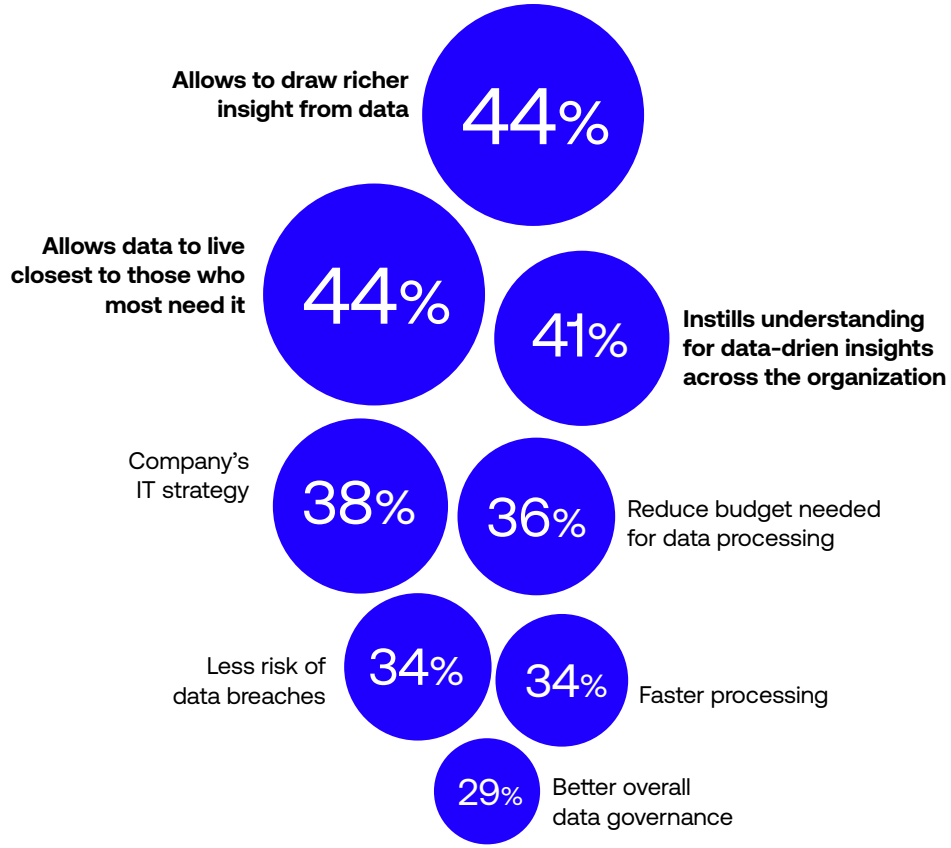


Figure 6: Digital Realty, *Global Data Insights Survey*, August 2024



Trend 4

Data drives the business agenda

Financial services businesses know the impact data can make, from operational improvements to driving customer engagement. As data drives business, business initiatives need to support that data with strategies for management, infrastructure, and location.

Being able to extract those impactful insights from data starts with a strategy for collecting data from across the organization into one location, as well as a plan for data hygiene to clean and structure the collected data.

Once harnessed, data drives the business agenda of a financial services company in several ways:

1. It can track the impact of new innovation or help businesses pinpoint areas where they could create new revenue streams.
2. It can make customer interactions or services offering more personalized, increasing engagement and retention.
3. It can improve forecasting, resulting in cost savings and better market positioning.

Today's AI technology can enhance each of these areas, giving financial services companies a competitive advantage.

Here's how financial services companies are using data to drive their business agenda.

Using AI to improve customer experiences

IT leaders expect that data-driven initiatives will increase the value they provide customers. **Over half (54%) believe that they can use their data-driven insights to increase customer satisfaction and improve customer experiences** — slightly more than 51% worldwide who say the same, showing the importance of customer relations in financial services.

Data can also help make their marketing efforts more effective by allowing them to send tailored messages and offers to their customers at the right time, through the right channels.

Additionally, IT leaders believe that the biggest value from executing their AI strategy will be to improve customer experiences. Not only can AI enhance the ability to extract insights from data, but AI can serve as those new business offerings — like a helpful banking chatbot or assistant, or using AI in fraud detection — to improve customer experiences, engagement, and retention.

What is the expected value from executing your AI strategy?

71%
Use AI insights to improve customer experiences

66%
Building AI capabilities into product or services

51%
Using AI to make busines operations more efficient

37%
Monetize data insights

13%
Create a competitive advantage

4%
Achieve regulatory compliance

Figure 7: Digital Realty, *Global Data Insights Survey*, August 2024



Using AI to foresee risk

In a time when one data breach can cost an organization an average of $4.45 million, financial services IT leaders believe data can decrease overall business risk.[1] The way they're using AI the most is to foresee risks in their organization or industry, and one of the top strategic outcomes their organization expects from their data-driven insights is to mitigate risks and cybersecurity breaches.

One way to mitigate risk in financial services is to use AI for customer onboarding. AI can analyze customer data and behavior to ensure that Know Your Customer (KYC) and Anti-Money Laundering (AML) compliance is met, and that new customers won't pose fraud risks.

Financial services companies can also use AI to enhance their cybersecurity efforts. AI can monitor the attack surface for suspicious activity and compare any activity against vast security event databases and logs faster and more efficiently than humans could. AI can also automatically investigate and respond to alerts, like phishing emails, and can help security teams improve by analyzing performance.

[1]IBM, *Cost of a Data Breach Report 2023*, 2023.

Which strategic outcome does your organization expect from data-driven insights in the next two years?

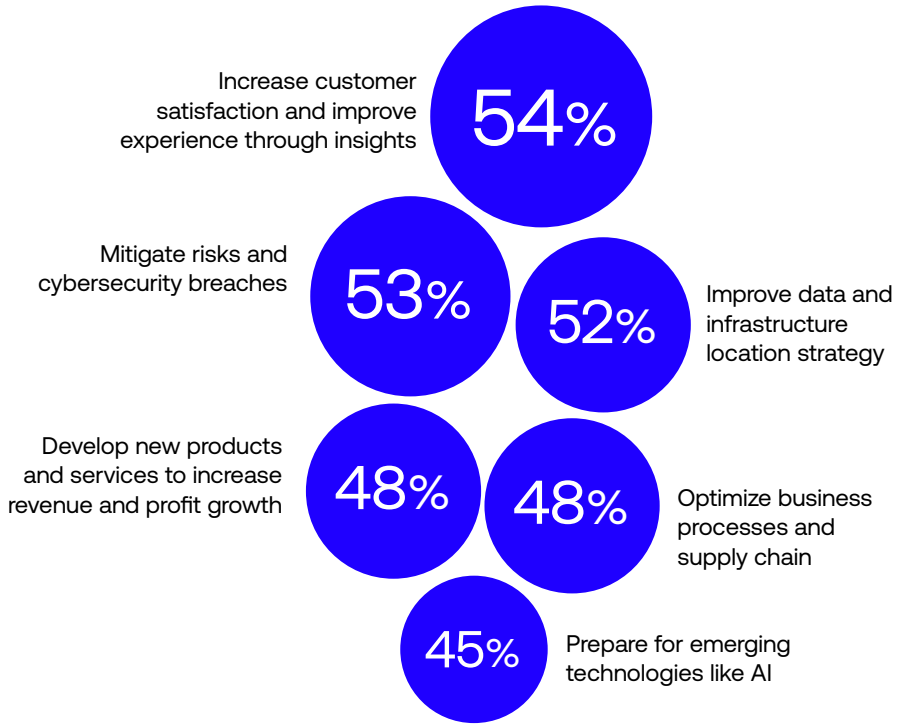


Figure 8: Digital Realty, *Global Data Insights Survey*, August 2024

How to leverage data and AI to fulfill enterprise goals

As IT leaders build their data strategies, they're looking to data center providers to help them. To realize their business goals, IT leaders need their data center providers to provide global coverage, capacity, and direct connectivity needs across major metros on a single data center platform. IT leaders using a single global platform scale quicker and with the right ecosystem of connectivity, data exchange becomes more seamless across users, networks, clouds, and IT service providers.

IT leaders need to plan ahead for budgetary investments to support these initiatives. In addition to being impacted by a lack of data infrastructure investment, the biggest impact on their IT infrastructure budgets will be building the systems, strategies, and approaches that allow them to analyze and draw insights from data — like new AI initiatives and the infrastructure to support it. However, budget allotment and leadership buy-in will only increase as IT leaders continue to demonstrate a return on investment with their data-driven and AI initiatives.





Build digital infrastructure foundations

for data and AI success

Enabling platform growth and AI-ready architecture requires a new digital infrastructure foundation. A data and AI-ready IT framework should support these four strategic IT foundations:



Network

As network traffic shifts, companies must establish the right network foundation to connect data, users, and applications. This network foundation must:

- Consolidate and localize traffic
- Segment and tier traffic
- Interconnect networks, cloud, and service providers
- Deploy, interconnect, and host SDN edge



Security

Against the backdrop of heightened cybersecurity threats, enterprises must secure controls and infrastructure in a distributed manner to ensure seamless global security. Enterprises need to secure, control, and protect infrastructure in order to:

- Implement ingress/egress control points
- Host IT and security stacks at ingress/egress points
- Securely, direct connect SaaS security/data services
- Enhance security posture to reduce vulnerability points



Compute

Given that most data remains outside of public cloud, enterprises require distributed and secure hybrid infrastructure that enables them to transact, process, and enable data and AI-driven insights. This distributed infrastructure should be designed to:

- Host analytics adjacent to network and data aggregation points
- Accelerate compute-intensive workloads with GPU/DPU
- Enable microservices with virtualization and containers
- Scale processing with clustering and high-performance interconnects



Data

Siloed data and distributed data workflows are nonperformant on legacy hardware. Data infrastructure that localizes data aggregation, staging, analytics, streaming, and management at critical global points of business is necessary for optimal AI-ready workflows. This infrastructure should:

- Host data adjacent to network and analytics aggregation points
- Optimize data exchange between users, things, networks, and clouds
- Enable real-time intelligence across distributed workflows locally and globally
- Create secure B2B data exchange to unlock new business opportunities

To be data and AI-ready means first solving your Hybrid IT journey. This means placing your digital infrastructure foundations and connecting them in a distributed, data-centric manner.

The PDx® Hybrid IT Journey

With our patented Pervasive Datacenter Architecture (PDX®) solution methodology and library, you can now colocate and interconnect digital infrastructure foundations locally to enable centers of data exchange.

- The PDx® solution library includes pre-validated use cases supporting the Hybrid IT journey. These use cases represent problems that enterprises need to solve on their Hybrid IT journeys.
- The PDx® solution library is organized around the implementation of four Hybrid IT initiatives: Work, Cloud, Data, and AI.

Spotlighting PDx® use cases

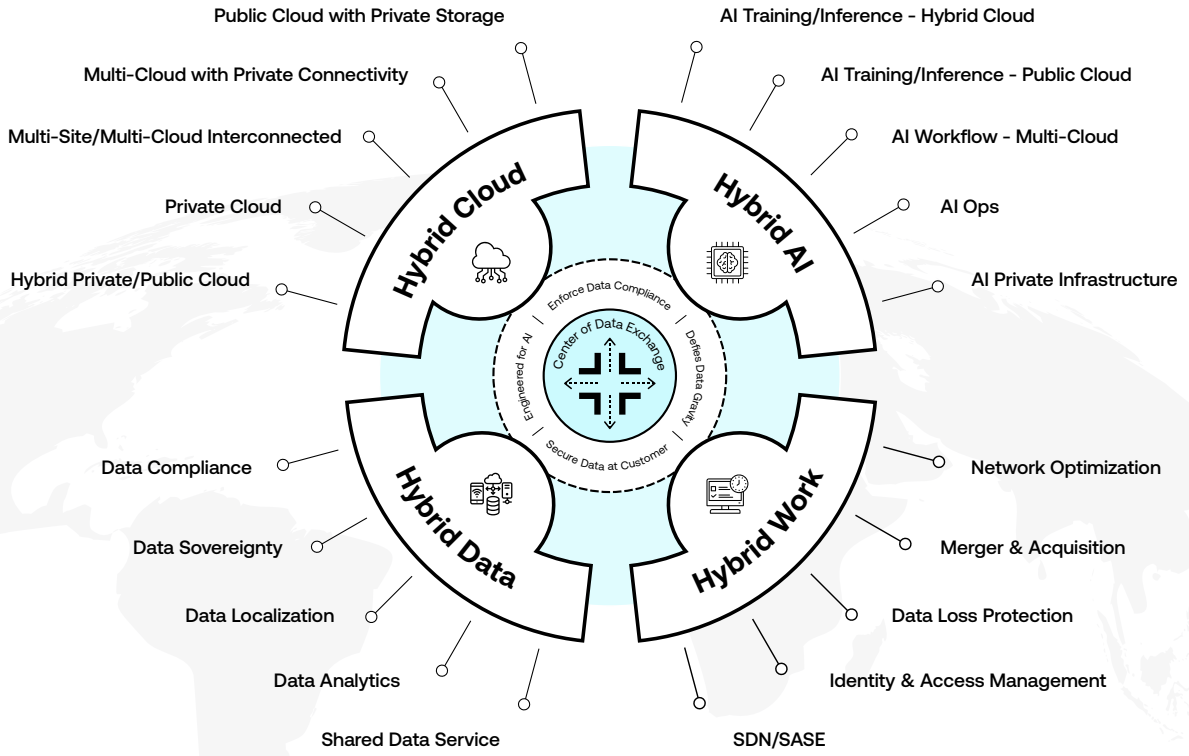


Figure 9: Digital Realty, Pervasive Datacenter Architecture®

Begin your data and AI-ready strategy

At your desk tomorrow

- Understand core business objectives
- Get your data house in order
- Future-proof IT architecture
- Pick the right partner

Get started with Digital Realty today and these four steps already listed here.

1. Request a strategy briefing

2. Engage in a solution workshop

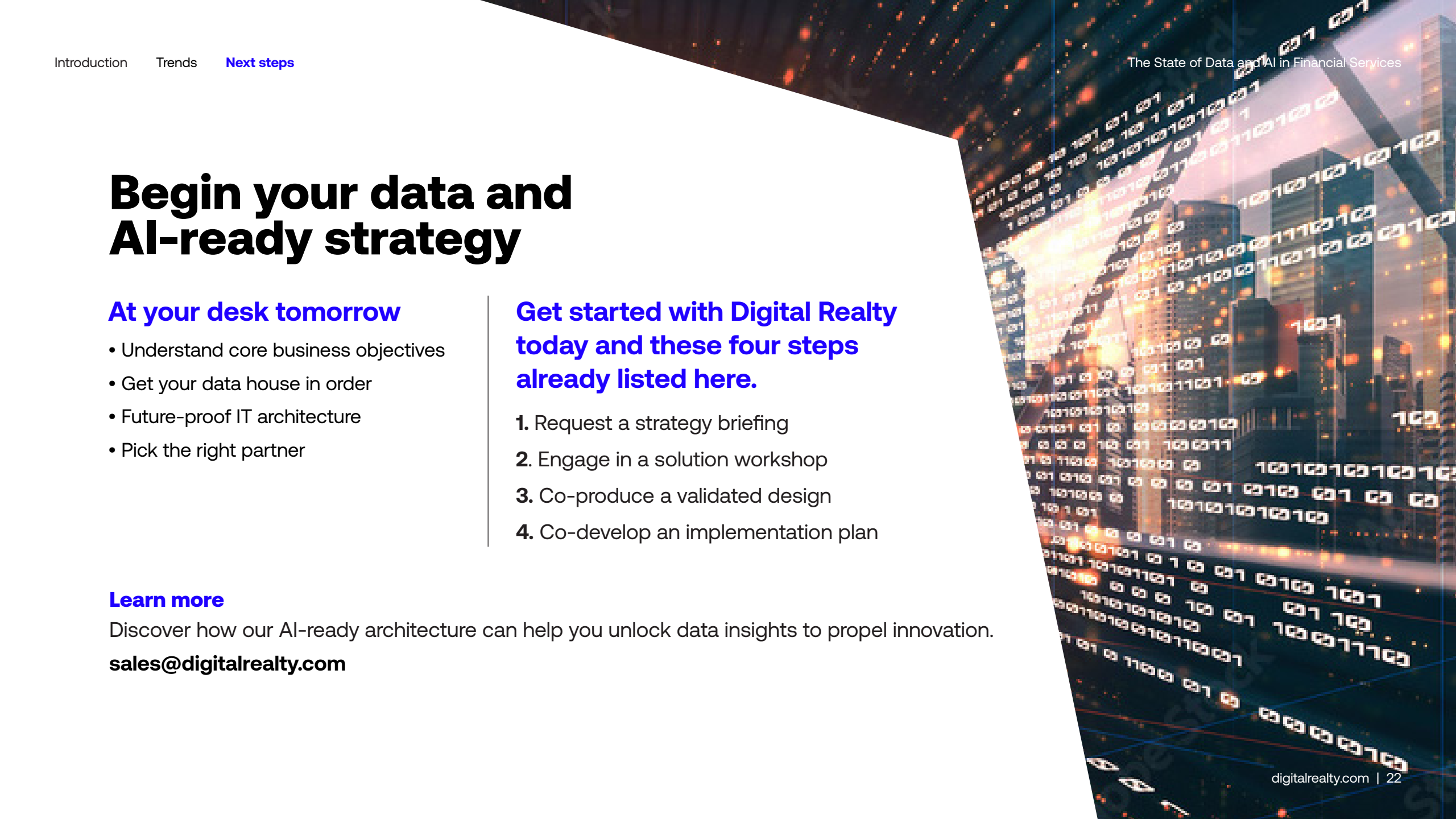
3. Co-produce a validated design

4. Co-develop an implementation plan

Learn more

Discover how our AI-ready architecture can help you unlock data insights to propel innovation.

sales@digitalrealty.com



About Digital Realty

Digital Realty brings companies and data together by delivering the full spectrum of data center, colocation, and interconnection solutions. PlatformDIGITAL®, the company's global data center platform, provides customers with a secure data meeting place and a proven Pervasive Datacenter Architecture (PDx®) solution methodology for powering innovation and efficiently managing Data Gravity challenges. Digital Realty gives its customers access to the connected data communities that matter to them with a global data center footprint of 300+ facilities in 50+ metros across 25+ countries on six continents.

To learn more about Digital Realty, please visit **digitalrealty.com** or follow us on **LinkedIn** and **X**.

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